

Grafana Panels Reference — Person 2 Module

AI-Powered Electronics Manufacturing Intelligence — 7 dashboards, 12 panels, built on electronics_manufacturing_db

Dashboard	Panel Name	SQL Query	Explanation
Production	Production Trends	SELECT production_date AS time, SUM(units_produced) AS units_produced, SUM(units_rejected) AS units_rejected FROM production GROUP BY production_date ORDER BY production_date;	Shows daily units produced vs units rejected as a time series. Helps spot whether output and rejects are trending up or down together.
	Total Production	SELECT SUM(units_produced) AS total_units FROM production;	Single stat showing the total number of units produced across the whole dataset. Gives a quick top-level output number.
Machine status	Machine Status Overview	SELECT status, COUNT(*) AS machine_count FROM machines GROUP BY status;	Counts how many machines fall into each status category (e.g. operational, maintenance). Gives a quick equipment health snapshot.
Quality	Defects Over Time	SELECT inspection_date AS time, SUM(defect_count) AS total_defects FROM quality GROUP BY inspection_date ORDER BY inspection_date;	Shows total defect count per inspection date. Helps spot spikes or trends in product defects over the monitored period.
	Defects by Type	SELECT defect_type, SUM(defect_count) AS total FROM quality GROUP BY defect_type ORDER BY total DESC;	Ranks defect types by total count. Identifies which defect category is the biggest quality problem.
Downtime	Downtime Over Time	SELECT downtime_start AS time, downtime_hours FROM downtime ORDER BY downtime_start;	Plots every downtime event by when it started and how long it lasted. Shows how frequently and severely machines stop running.
	Top Downtime Reasons	SELECT downtime_reason, SUM(downtime_hours) AS total_hours FROM downtime GROUP BY downtime_reason ORDER BY total_hours DESC;	Ranks downtime causes by total hours lost. Highlights the single biggest driver of machine downtime.
Maintenance	Recent Maintenance Log	SELECT maintenance_date, maintenance_type, maintenance_status, downtime_hours, maintenance_cost FROM maintenance ORDER BY maintenance_date DESC;	Row-by-row history of maintenance events with type, status, downtime, and cost. Acts as a searchable maintenance record.
Sensors	Temperature Trend	SELECT recorded_at AS time, sensor_value FROM sensors WHERE sensor_type = 'Temperature' ORDER BY recorded_at;	Time series of temperature sensor readings. Has an alert rule attached that flags readings above a set threshold.

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	Vibration Trend	SELECT recorded_at AS time, sensor_value FROM sensors WHERE sensor_type = 'Vibration' ORDER BY recorded_at;	Time series of vibration sensor readings. Helps catch unusual mechanical vibration patterns on machines.
KPIs	Defect Rate (%)	SELECT SUM(units_rejected)/SUM(units_produced)*100 AS defect_rate FROM production;	Single stat showing the overall percentage of produced units that were rejected. A core quality KPI.
	Total Downtime Hours	SELECT SUM(downtime_hours) AS total_downtime FROM downtime;	Single stat showing total downtime hours across all machines and events. A core reliability KPI.